

RaceDay Requirements Traceability

Purpose

This document maps the main RaceDay Part 1 requirements to the technical deliverables completed in the repository.

Traceability Matrix

RaceDay Requirement	ERD	SQL Database	API Plan	Evidence
User accounts and roles	User entity with Role	User table and Role constraint	Register and Login endpoints	Complete
Organisers manage events	User → Event relationship	OrganiserID foreign key	Event POST, PUT and DELETE endpoints	Complete
Participants enter events	User → Enrolment relationship	ParticipantID foreign key	Enrolment POST endpoint	Complete
Participants select categories	Event → Category and Category → Enrolment	CategoryID and EventID foreign keys	Enrolment request includes categoryId	Complete
Organisers view enrolments	Event → Enrolment relationship	Enrolment foreign keys	Event enrolment endpoint	Complete
Organisers capture results	Enrolment → Result relationship	Result table and foreign key	Result POST and PUT endpoints	Complete
Participants view results	Enrolment → Result relationship	Result linked to Enrolment	Participant results endpoint	Complete
Event types are supported	EventType → Event relationship	EventTypeID foreign key	Event creation includes	Complete

RaceDay Requirement	ERD	SQL Database	API Plan	Evidence
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Part 1 Deliverables

ERD

The ERD documents the entities, attributes, primary keys, foreign keys, relationships and cardinality.

SQL Database

The SQL script implements the ERD as a SQL Server database and includes constraints and realistic sample data.

API Endpoint Plan

The API plan defines the future RESTful API required to interact with the RaceDay data and functionality.

GitHub Actions

The CI workflow validates the required Part 1 repository files.

Conclusion

The RaceDay ERD, SQL database and API Endpoint Plan were designed to support the same system requirements and form a consistent technical foundation for the later implementation phases.